Section A – Answer any 2 out of 3 questions in section A

Question 1

a) There are two identical generators, each of 100 kW capacity and availability of 95%. Assume the reliability criterion requires that the probability to meet peak demand is above 92%, and the coincidence factor of an infinite number of households is 0.444. The peak demand of each individual household is 5 kW.

(i) how many households can be supplied by these two generators?

[4]

(ii) how many additional households can be supplied if a 100 kW wind farm with 50% availability is connected to the system?

[4]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Enhanced Frequency Response (EFR) in the UK. EFR reduces the need for the regular frequency response (R), following this rule :

$$\left(\frac{H}{f_0} - \mathrm{EFR} \cdot \frac{\mathrm{T_{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \cdot \mathrm{T_d}}{4 \cdot \Delta f_{\max}}$$

Nomenclature

H: system level of inertia (Units: MW · s).
R: amount of frequency response scheduled (Units: MW).
EFR: amount of EFR scheduled (Units: MW).
P_{L}: value of the generation outage we want to secure against (Units: MW).
f_0: nominal electric frequency of the power grid (Units: Hz).
$\Delta f_{\max}$: maximum admissible frequency deviation to avoid load shedding (Units: Hz).
$\mathrm{T_d}$: Delivery-time of frequency response (Units: s).
$\mathrm{T_{EFR}}$: Delivery-time of EFR (Units: s).

Assumptions:
1. The system level of inertia of 150GWs, i.e. H = 150 GW-s,
2. The cost of frequency response: $5/MW,
3. The largest generation unit in the UK: 1.8 GW,
4. Other numerical values: f0 = 50 Hz, Δfmax = 0.8 Hz, Td = 10 s, TEFR = 0.5 s.

If there is a bid of 200MW of EFR for £20/MW, should the system operator buy it to achieve the minimum cost?

[6]

c) In a balanced three-phase star-connected system, as shown in Figure 1, write down the expressions for current through the neutral connection I_n as functions of voltage magnitudes and phase angles.

[6]

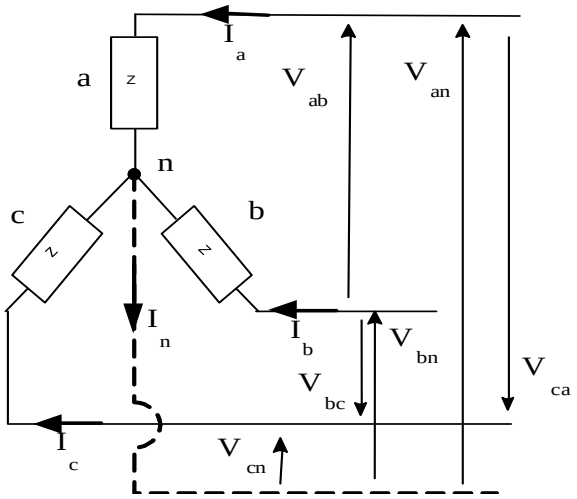


Figure 1 Balanced three-phase star-connected system

Question 2

a) What are the principles in selecting the base value for per-unit analysis in power systems?

[3]

b) Consider a high voltage transmission circuit in Figure 2.1. Assume that voltages (magnitudes and phase angles) are known at both sending and receiving ends.

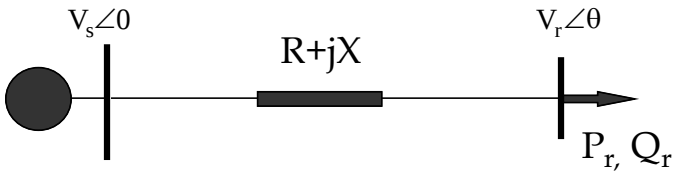


Figure 2.1 Transmission circuit with voltages specified at the both sending and receiving ends

(i) Write down the expressions for sending end active and reactive powers as functions of voltage magnitudes and phase angles.

[3]

(ii) Based on these expressions and assuming R>>X, show that active power flow requires a difference in voltage magnitude and that reactive power flow requires a difference in phase angle.

[3]

c) There is a three-phase system as shown in Figure 2.2. A generator with a capacity of 60 MVA is connected to a 15 kV bus, which is fed into a 11 kV substation. The load receives active power 36 MW and reactive power 27 MVA. Calculate the voltage at the receiving end (V_r) and the complex power (P_s+jQ_s) at the sending end given $S_B = 50$ MVA.

[11]

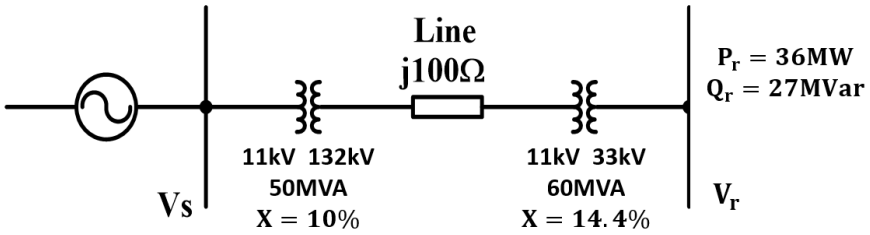


Figure 2.2 Three-phase system is supplied by a generator

Question 3

A four-bus power network is presented in Figure 3 below. Data relevant for the load flow analysis on this system are given in per-unit values.

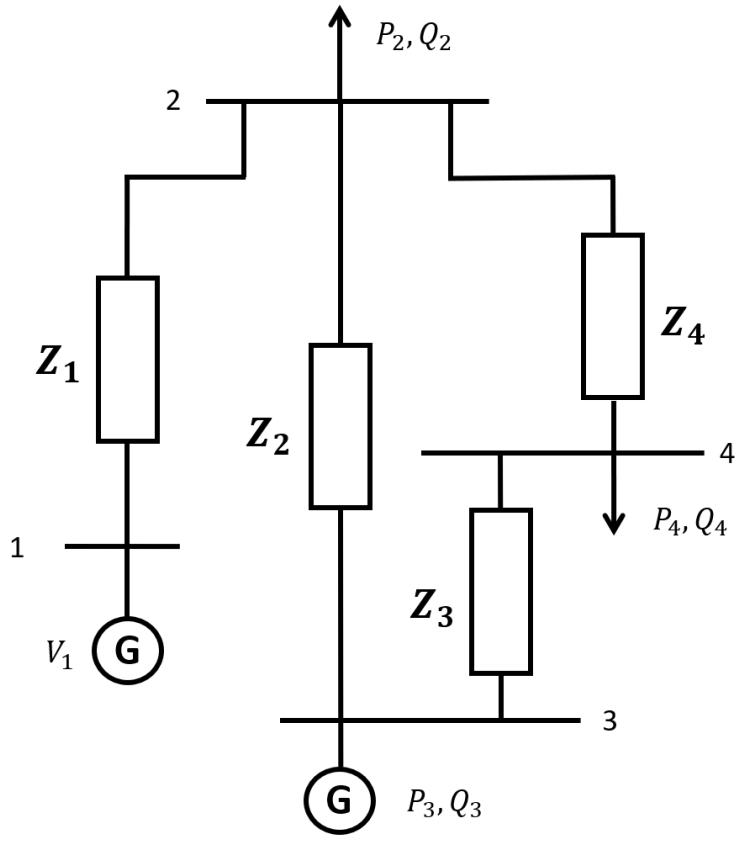


Figure 3. A four-bus power network

a) Given Z_1, Z_2, Z_3 and Z_4, form the bus admittance matrix

[6]

b) Assuming the per unit values $|V_1| = 1$, $|V_3| = 1.1$, $P_3 = 1.1$, $P_2 = 0.95$, $Q_2 = 1$, $P_4 = 1.05$, $Q_4 = 1$, $Z_1 = 0.25j$, $Z_2 = 0.25j$, $Z_3 = 0.2j$ and $Z_4 = 0.2j$, perform two iterations of the Gauss-Seidel load flow method.

[10]

c) List the possible short- and long-term options if the power flow across a transmission line exceeds its rated capacity.

[4]

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) Justify or rectify the following statements with proper explanation:

(i) The bus impedance matrix used for fault analysis can be obtained by inverting the bus admittance matrix that is used for power flow analysis.

[2]

(ii) The flux linkage in a closed circuit with finite electromotive force and finite resistance cannot change instantaneously.

[2]

(iii) It is completely unreasonable to assume the pre-fault line currents to be zero in order to simplify fault calculations.

[2]

(iv) The short circuit level of a power system reduces when conventional synchronous generation is replaced by wind and solar photovoltaic generation.

[2]

b) Consider a power system with a total number of buses N_b. A 3-phase fault occurs at bus k with a fault impedance Z_f. Derive the expression for the following in terms of the pre-fault voltage V_i^0 at bus i, appropriate elements of the bus impedance matrix $\boldsymbol{Z_{BUS}}$ and fault impedance Z_f. Assume that the row/column number of $\boldsymbol{Z_{BUS}}$ denotes the bus number and that Z_{mn} denotes the element in the mth row and nth column of $\boldsymbol{Z_{BUS}}$.

(i) Fault current I_f

[4]

(ii) Voltage V_i^f at any bus i during the fault

[2]

c) The bus impedance matrix ($\boldsymbol{Z_{BUS}}$) of a power system with 3 buses is given below where the row/column number of $\boldsymbol{Z_{BUS}}$ denotes the bus number.

$$\boldsymbol{Z_{BUS}} = \begin{bmatrix} j0.15 & j0.08 & j0.05 \\ j0.08 & j0.13 & j0.04 \\ j0.05 & j0.04 & j0.14 \end{bmatrix} \text{p.u.}$$

Consider a 3-phase fault at bus 2 with zero fault impedance, neglect the pre-fault currents and assume the pre-fault voltages at all the buses to be 1.0 p.u.

(i) Calculate the fault current (p.u.).

[2]

(ii) Calculate the voltage (in p.u) at each bus during the fault.

[3]

(iii) Find the Thevenin equivalent impedance of the power system at bus 1 and 3.

[1]

Question 5

a) The phase currents drawn by a star connected load with an ungrounded neutral point are $I_A = 1.05\angle 90°$ p.u. and $I_C = 0.99\angle 20°$ p.u. Calculate the magnitude (in p.u.) and phase angle (in degrees) of the positive- (A-B-C), negative- (A-C-B) and zero-sequence components of the load current.

[6]

b) For a line-to-ground (LG) fault in phase A, show that the magnitude of the fault current $|I_f|$ (in phase domain) is given by:

$$|I_f| = \left|\frac{3V_f}{Z_1 + Z_2 + Z_0 + 3Z_f}\right|$$

where Z_1, Z_2, Z_0 are the equivalent positive-, negative- and zero-sequence impedance, Z_f is the fault impedance and V_f is the pre-fault phase voltage.

[4]

c) Two star-connected generators are connected in parallel to supply a static load through a line. The neutral point of one of the two generators is grounded through a reactor with $X_N = j0.1$ p.u. The neutral point of the other generator is left ungrounded. The values of the sequence reactance (in p.u.) of the generators and the line are given in Table 5.1.

Table 5.1: Sequence reactance of the generators and the line

	X_1	X_2	X_0
Generator	$j0.2$	$j0.18$	$j0.05$
Line	$j0.07$	$j0.07$	$j0.12$

A line-to-ground (LG) fault occurs on phase A at the bus where the static load is connected. The fault impedance is $Z_f = j0.03$ p.u. Assume the pre-fault voltages to be 1.0 p.u. and neglect the pre-fault current.

(i) Draw the three sequence networks for the LG fault condition showing the sequence reactance for individual components.

[2]

(ii) Calculate the LG fault current (in p.u.).

[4]

(iii) If the LG fault current was to be limited to half of that under the above situation, what value of neutral grounding reactance X_N (in p.u.) would be necessary?

[2]

(iv) Find the minimum value of the neutral grounding reactance X_N (in p.u.) to ensure that the fault current for an LG fault at the terminals of the generator is not more than the corresponding 3-phase fault current for zero fault impedance.

[2]

Question 6

a) Define transient stability of power systems and state the two factors that primarily affect the transient stability.

[2]

b) A coal-fired power plant with round-rotor synchronous generators supplies a large power system through a transmission line. The voltages at the two ends of the line are regulated at 1.0 p.u. The reactance of the transmission line is 0.5 p.u. and its resistance and shunt charging are negligibly small. The resistance and reactance of the synchronous generators are also negligible. Determine the maximum power (in p.u.) that can be transferred across the line in steady-state for the following conditions and sketch the corresponding power angle characteristics in each case.

(i) without any line compensation.

[2]

(ii) with 35% series compensation installed on the transmission line.

[2]

(iii) with ideal phase angle compensation of -15° at one end without changing the effective voltage at that end.

[2]

(iv) with an ideal mid-point shunt reactive compensation installed exactly halfway through the length of the transmission line to maintain the mid-point voltage at 1.0 p.u.

[2]

(v) for case (iv), calculate the reactive power (in p.u.) capacity required for the shunt compensator.

[2]

c) Consider the same context as in part (a) of this question. At a given point, the system is operating at 50% of its steady-state stability limit without any line compensation. Assuming the power output from the generators to be zero during a fault on the transmission line calculate the critical clearing angle (in degrees) for the following situations:

(i) without any line compensation.

[4]

(ii) with 35% series compensation but the generator output (i.e. power transfer through the line) remaining the same as in part (i).

[2]

(iii) what is the power transfer (in p.u.) across the line with 35% series compensation if the critical clearing angle is the same as that in part (i)?

[2]